GNBF5020

Lecture 7 · Exam Focus & Learn-from-Zero Guide

Genomic Applications in Biomedical Research — the block he flagged, the message he called 'very clear and simple', the four traps, the numbers, and ten concept chains that take you from zero.

GNBF5020 · Introduction to Molecular Biology and Genetics

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MSc in Genomics and Bioinformatics · GNBF5020 · 3 units · Term 1, Saturday 16:15–18:30

Lecture No. 7 — Genomic Applications in Biomedical Research · 2 h 27 min · 7 Nov 2026

Assessment weighting (published): Assignments 30% · Mid-term 30% · Final 40%

Source: the 2 h 27 min recording and its 42-slide deck, transcribed in full, section by section

[ENGLISH ONLY] ALL EXAM ANSWERS MUST BE IN ENGLISH. Dr. Wong's own words, from Lecture 1: *"every assignments and exam should be done in English, even this is a Chinese University of Hong Kong."* And again in Lecture 4: *"you have to answer in English, although this is a CUHK, haha, okay."* He said he gave zero marks to students who answered in Chinese last year. In the Traditional Chinese edition of this book every warning box is in Chinese, but **all gene, protein, pathway, method and drug names stay in English**.

[HOW TO USE THIS BOOK] Two halves, two jobs. Part A is the exam half: the block Dr. Wong flagged by name, the message he called 'very clear and simple', the four traps, the numbers and thresholds, the eighteen discriminator pairs, the method-selection table, and a fifteen-question self-test with a full answer key — read it the night before the paper. Part B is the learning half: ten concept chains that take you from zero clinical genomics to being able to *explain* the lecture rather than recognise it — work through it in order, with a pen in hand. Five figure plates summarise the pipeline, the biomarker-versus-CDx distinction, the prognostic-versus-predictive split, the validation ladder and the five-day plan.

繁體中文版 Traditional Chinese edition: GNBF5020_第七課考試重點與學習指南_繁體中文.pdf

Video guides

Twelve terms in this book carry the most exam weight, and each one has a short video guide: **English narration, Traditional Chinese subtitles**, about a minute each. The MP4 files live in the project folder **_video/assets/** - the full path is printed next to every link below, so you can still find a file after this PDF has been moved. **_video/index_lec7.html** plays all Twelve on one page.

Video	Term	Chapter	Video file
L01	**Biomarker vs companion diagnostic**	Lecture 7 · Genomic Applications in Biomedical Research	_video/assets/L01_Biomarker_vs_companion_diagnostic.mp4
L02	**Prognostic vs predictive biomarker**	Lecture 7 · Genomic Applications in Biomedical Research	_video/assets/L02_Prognostic_vs_predictive_biomarker.mp4
L03	**Driver vs passenger gene**	Lecture 7 · Genomic Applications in Biomedical Research	_video/assets/L03_Driver_vs_passenger_gene.mp4
L04	**Pathway enrichment analysis**	Lecture 7 · Genomic Applications in Biomedical Research	_video/assets/L04_Pathway_enrichment_analysis.mp4
L05	**Real-time PCR vs droplet digital PCR**	Lecture 7 · Genomic Applications in Biomedical Research	_video/assets/L05_Real_time_PCR_vs_droplet_digital_PCR.mp4
L06	**MammaPrint vs Oncotype DX**	Lecture 7 · Genomic Applications in Biomedical Research	_video/assets/L06_MammaPrint_vs_Oncotype_DX.mp4
L07	**Recurrence Score and chemotherapy benefit**	Lecture 7 · Genomic Applications in Biomedical Research	_video/assets/L07_Recurrence_Score_and_chemotherapy_benefit.mp4
L08	**Tumour tissue vs tumour-adjacent tissue**	Lecture 7 · Genomic Applications in Biomedical Research	_video/assets/L08_Tumour_tissue_vs_tumour_adjacent_tissue.mp4
L09	**Analytical validation**	Lecture 7 · Genomic Applications in Biomedical Research	_video/assets/L09_Analytical_validation.mp4
L10	**Autoantibody vs anti-drug antibody**	Lecture 7 · Genomic Applications in Biomedical Research	_video/assets/L10_Autoantibody_vs_anti_drug_antibody.mp4
L11	**Enrichment design vs all-comer design**	Lecture 7 · Genomic Applications in Biomedical Research	_video/assets/L11_Enrichment_design_vs_all_comer_design.mp4
L12	**CRISPR knock-out vs knock-in**	Lecture 7 · Genomic Applications in Biomedical Research	_video/assets/L12_CRISPR_knock_out_vs_knock_in.mp4

Open the video index page ->

§0 How to use this book

This book does two jobs at once, and they are deliberately interleaved.

Part A is the exam half. It is built from what Dr. Wong actually said about the exam, not from a general sense of what might be examinable. Where he named a question, flagged a trap, or told you what the marks are for, it is quoted. Where he did not, it is left out. You should be able to read Part A the night before the paper and walk in with a short list.

Part B is the learning half. It assumes you are starting from zero on clinical genomics and takes you to being able to *explain* the lecture rather than recognise it. It is organised as ten concept chains. A chain is not a list of facts; it is a line of reasoning you can reproduce out loud under exam pressure.

The two halves answer different questions, and you should read them differently:

	Part A — Exam Focus	Part B — Learn from Zero
Question it answers	*What will be on the paper?*	*How do I understand this at all?*
Read it by	reading, then drilling from memory	working through, in order, with pen in hand
Success looks like	you can recite the numbers and the pairs	you can explain why each test exists
Time needed	3–4 hours total, in pieces	8–12 hours, spread over 5 days

One warning before you start. Dr. Wong's own remark from the earlier lectures sets the standard for everything in this book:

> *"Use simple sentences, short but clear sentence to make the description. That's the key to the exam."*

The marks are not for vocabulary. They are for a clear, mechanistically correct, short answer written in English. Every section below is written to be imitated in that style.

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§1 The lecture on one page

The lecture is built as **seven moves**, and it never leaves the world of clinical genomics: it starts with a table of genes and ends with a patient who either gets the right drug or does not.

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1 OMICS DATA → A PATHWAY slides 3–5 pathway enrichment analysis (PEA)
RNA-Seq / microarray → DEG list → g:Profiler, GSEA, Cytoscape
↓
2 A PATHWAY → A REAL TARGET slide 6 driver vs passenger; CNV + PEA
upregulated gene + copy-number change + enriched pathway → driver gene
↓
3 A TARGET → A DRUG slides 6–8 small molecule vs antibody; tox; IND; Phase 1/2/3
↓
4 A DRUG → A TEST THAT FINDS THE PATIENT slides 7–10 biomarker vs companion diagnostic (CDx)
HER2 as biomarker; HER2 IHC or qPCR as the CDx; $200,000 per patient enrolled
↓
5 A TEST → A SCORE slides 12–18 MammaPrint (70 genes) and Oncotype DX (21 genes)
↓
6 A SCORE → A TREATMENT DECISION slides 19–22 prognostic vs predictive; chemo only in high RS
↓
7 A DECISION → A PERSON slides 23–25 adjacent tissue; pharmacogenomics; FDA labelling
↓
…and the tools that make the whole thing work:
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slides 26–42 biomarker development, analytical validation, clinical validation,
clinical trial design, CRISPR, organoids, liver-chip
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Three things are worth noticing about that shape.

First, **the lecture is a pipeline, not a topic list.** Almost every slide is a box in the same flow: data → biology → drug → test → patient. If you can draw the pipeline on a blank page and put the right thing in each box, you can answer almost any question the paper asks.

Second, **the middle of the lecture is where the marks live.** Dr. Wong spent roughly 55 of the 147 minutes on the two breast-cancer signatures and the chemotherapy-benefit trial. That is the densest, most examinable part of the lecture, and it is also the part with the most numbers on the slides.

Third, **the back half is the part he flagged as the exam focus** (see A1.1). He said he would spend time on *"how we actually developed a new biomarkers"* because he knows some of you work in biomarker or IVD companies. The biomarker-development pipeline — analytical validation, clinical validity, clinical utility — is not an aside in this lecture. It is the point of the second half.

Figure L7 — The lecture is one pipeline: from a gene list to a patient



Figure L7. The seven moves of the lecture, in the order he teaches them: omics data to a pathway, a pathway to a real target, a target to a drug, a drug to a test that finds the patient, a test to a score, a score to a treatment decision, and a decision to a person. Figure authored at 6.5 in print width and placed 1:1 on this page.

How to read this plate: start at move 1 and work down. Each move takes the output of the one above it and turns it into the input of the next. If you can draw this on a blank page, you can place almost any fact the paper asks for inside it.

§2 The frame for everything: nothing is a biomarker until it changes a decision

Dr. Wong opened by placing this lecture explicitly at the end of the molecular biology block:

> *"Okay, good afternoon class. Uh today is the last lecture about the molecular biology. So, starting from next week, we are going to talk about the uh genetics."*

And then he said why the lecture is worth an afternoon:

> *"So, uh I think today is more related to uh the application of uh all this genomic technology and uh that may be more interesting uh to some of you… And uh if you have uh read my uh lecture uh materials uh today uh I will spend some time to uh to uh discuss uh how we actually developed a new biomarkers. So, not only from the uh technology background but also uh how how you prepare uh your data for the regulatory approval, yeah. I think that can be more interesting to some student."*

2.1 The single idea that runs through the whole lecture

Here is the sentence that holds the lecture together. He said it near the end, about the tumour-adjacent tissue study, but it is the thesis of the entire 147 minutes:

> *"The key messages that I would like to deliver is, don't only look at the cancer tissue, but also the adjacent tissue in your future bioinformatic analysis. So this is the only message that I would like to deliver here. Very clear and simple message."*

Read that twice, because it works at two levels. At the level of the data, it says: the biology of a cancer is not contained in the cancer cells; the tissue next to the tumour carries prognostic information of its own. At the level of the whole course, it says: a measurement only becomes medicine when you know what to do with it.

That second reading is the frame for everything that follows:

- A **gene list** is not a target until pathway enrichment tells you which biology it belongs to.
- A **target** is not a drug until you have a molecule, a mechanism of action and toxicology.
- A **drug** is not useful until you can find the patients who will respond.
- A **biomarker** is not a test until it is analytically validated, clinically valid, and — the last and hardest step — shown to improve an outcome.
- A **signature** is not a decision rule until you know what the numbers mean for the patient's treatment.

Every block of the lecture is one more rung on that ladder. Learn the ladder and the lecture is memorable, because it is one idea, not forty.

2.2 The two words that carry the most marks

If you only learn two terms from this lecture, learn these two, because they are the terms he told you to understand and the terms that distinguish a good answer from a correct-but-thin one:

Term	His definition, in his own words	What it is NOT
Prognostic	tells you how the patient will do, whatever treatment you give	it does **not** tell you which drug to give
Predictive	tells you whether *this* patient will benefit from *this* drug	it does **not** tell you how well the patient will do in general

He flagged them explicitly:

> *"And uh there's a two terms uh that uh uh I think you need to understand, uh because uh when you read the research article, uh you are always come across with this term: prognostic uh markers and also the predictive markers."*

Almost every other pair in this lecture (biomarker vs CDx, analytic vs clinical validation, driver vs passenger, hot vs cold tumour, qPCR vs ddPCR) is a version of the same discipline: say which of the two things you mean, and never let them blur.

2.3 Self-check

Before you read further, answer these three in one sentence each. If you cannot, Part B is for you.

1	What is the difference between a biomarker and a companion diagnostic?
2	A patient has an Oncotype DX Recurrence Score of 22. Is that score prognostic, predictive, or both?
3	Name the three things a laboratory assay must have before it can be used to pick patients for a trial.

PART A — EXAM FOCUS

What Dr. Wong actually said about the exam: the block he flagged by name, the message he called 'very clear and simple', the four traps, the numbers, the eighteen pairs, the answer skeleton, and the one thing he told you not to memorise.

A1 What he actually said about the exam

This section contains only what Dr. Wong actually said in this recording. He was not, in this lecture, as explicit about the exam as he was in Lecture 6; there is **no question he named by name** in this lecture. What he did do was flag a *block* of the lecture as the part most likely to matter, give one message he wanted you to carry away, and tell you one thing not to memorise. Those are the three signals, and they are what this section is built from.

A1.1 The block he flagged for you ⚑

> *"So I introduce, uh, different biomarkers and I think, uh, another logical, uh, question is, uh, how we uh, develop, uh, these biomarkers. And I think this may be uh, more relevance, uh, to uh, you guys, uh, because I know that, uh, some of you are working in uh, biomarkers company or IVD company. And, uh, I think the field the next few slide may be very uh, interesting, uh, to you guys."*

What this means for you. The biomarker-development block — slides 26 to 42, the last ~45 minutes of the lecture — is the part he pointed at. It contains:

- the FDA's own development framework (discovery → validation → utility);
- **analytical validation** — sensitivity, specificity, dynamic range, reproducibility, matrix effect;
- **clinical validation** — clinical validity and clinical utility, and the ROC curve;
- the **team** a biomarker needs (scientists, business development, medical writers, QA/GLP, regulatory affairs, project managers);
- **clinical trial design** — the enrichment design and the all-comer design.

If you are revising one block of this lecture, revise this one. The named exam questions in Part A (A2) are all drawn from it.

A1.2 The message he called "very clear and simple" ⚑

> *"But anyway, don't be scared by all these figures, yeah. The key messages that I would like to deliver is, don't only look at the cancer tissue, but also the adjacent tissue in your future bioinformatic analysis. So this is the only message that I would like to deliver here. Very clear and simple message."*

Three things are packed into this sentence and all three are examinable:

1	**Don't be scared by all these figures.** The lecture is dense; he is telling you not to panic at the volume, and to pick out the biology.
2	**Look at the adjacent tissue too.** A specific, checkable methodological instruction — in your own future bioinformatics work, profile the normal tissue next to the tumour, not only the tumour.
3	**It is a bioinformatics message, not only a clinical one.** He said *"in your future bioinformatic analysis"* — this is guidance for the analyses you will do, not only for the patients you will treat.

The study behind it: in breast cancer, the gene signature in **histologically normal tissue 2 cm from the tumour edge** carries its own prognostic information. Patients whose *adjacent* tissue carried an **"active"** signature had worse survival; those with an **"inactive"** signature did better — and he describes the "inactive" signature as enriched for **anti-tumour immunity** genes. The microenvironment, not just the tumour, is doing work.

[A CORRECTION] The deck's citation for this study does not exist, and the published paper's own axis runs the other way: in the source paper the **Active** subtype is the inflammatory, fibrosis-and-macrophage one, while **Inactive** is defined by cell adhesion and epithelial differentiation. **In the exam, say what he taught**; know the paper separately. See **C.2 #4 and #5**.

A1.3 The one thing he told you NOT to memorise ⚑

> *"So this is the cancer genes included in this structure and don't memorize all this genes. ID is not necessary. But you just need to understand all these genes are really, really involved in the cancer biology, like the proliferation and also the signaling involve in estrangement and HER2… And definitely some genes are involved in the cancer cell invasions. So for example, for MMP, so this is a metallo protein, the enzyme that digest the extracellular matrix."*

This is a direct release from study, and it is worth taking. You do **not** need to memorise the gene IDs of the 21-gene panel. You need to know the **five functional groups** they belong to:

Group	Genes on the slide	What the group does
Proliferation	Ki-67, STK15, Survivin, Cyclin B1, MYBL2	how fast the cells divide
Estrogen signalling	ER, PR, Bcl2, SCUBE2	the hormone pathway that drives ER+ breast cancer
HER2 signalling	GRB7, HER2	the receptor tyrosine kinase pathway
Invasion	ST3 (MMP11), CTSL2	breaking out of the tissue — matrix metalloproteinases digest extracellular matrix
Other	GSTM1, BAG1, CD68	other contributing processes

Say the **group and its meaning**, and you have the mark. Say "STK15" when asked what matters, and you have wasted the point.

A1.4 The lecture is the last of the molecular biology block ⚑

> *"Today is the last lecture about the molecular biology. So, starting from next week, we are going to talk about the genetics."*

If your exam paper has a genetics section, it comes from the lectures **after** this one, which are not in this book. What this tells you is where the boundary of the molecular biology module sits: this lecture closes it.

A1.5 His standing principle

Unchanged from the earlier lectures, and still the rule that decides your marks:

> *"Use simple sentences, short but clear sentence to make the description. That's the key to the exam."*

A1.6 What he said about *not* worrying

He explicitly released you from the volume of the lecture:

> *"Don't be scared by all these figures."*

And, in the same spirit, he said of the gene list:

> *"don't memorize all this genes. ID is not necessary."*

A2 The traps he named, in exam-ready form

There are **four** traps in this lecture. Each one is a place where a plausible answer is actually wrong, and each one is worth writing a full sentence about.

Trap 1 — The biomarker is not the companion diagnostic ⚑ (the named one)

This is the first thing he taught in the biomarker block, and it is the one most likely to be worth marks.

> *"So there's a two-step concept. One is the biomarker. So this is a single gene or signature, but the other is the test the method that you test the level of your biomarkers."*

The two-step concept, written the way to write it:

	What it is	Example from the lecture
The biomarker	the biological entity itself — a gene, a protein, or a whole expression signature	**HER2** overexpression; the 21-gene expression signature
The companion diagnostic (CDx)	the *assay* that measures the biomarker in a clinic	**HER2 immunohistochemistry (IHC)** or **HER2 qPCR** — the method used to select the patients

Both are needed and they are not the same thing. A biomarker without a CDx is a biological observation nobody can act on; a CDx without a validated biomarker is an instrument measuring nothing meaningful.

The CDx is not optional either — it is developed **alongside** the drug:

> *"The key message that I want to deliver is um along the drug uh development, you also need to develop a biomarker assay so that it can help you to select the right patients to be recruited for your clinical trial. So, we call this is a companion uh diagnostic uh CDX."*

Trap 2 — Prognostic is not predictive ⚑

> *"If you… always come across with this term: prognostic uh markers and also the predictive markers."*

	Prognostic	Predictive
Question it answers	How will this patient do?	Will this patient benefit from *this* treatment?
Depends on the therapy?	No — outcome regardless of which therapy is given	Yes — outcome **relative to** a specific therapy
Measured by	survival / recurrence independent of treatment arm	a **treatment-by-biomarker interaction** (does the biomarker separate responders from non-responders *within* a trial?)
Example	the Recurrence Score as an independent factor in multivariate analysis	the Recurrence Score separating who benefits from adjuvant chemotherapy

The exam-safe sentence:

> *"A prognostic biomarker indicates the patient's likely outcome irrespective of treatment; a predictive biomarker indicates differential benefit from a specific therapy, and it can only be demonstrated in a trial where the biomarker is compared across treatment arms."*

The second half of that sentence is the part that earns the mark: **a predictive claim requires a treatment arm.** A biomarker that is prognostic only cannot be used to justify giving a drug to one group and withholding it from another.

Trap 3 — Prognostic ≠ predictive, applied to the two breast-cancer tests

This is where Trap 2 gets applied, and it is the classic exam question.

- **MammaPrint (70 genes)** — a **prognostic** assay. It reports high risk / low risk of distant recurrence. It does not, on the evidence shown, tell you who benefits from chemotherapy.
- **Oncotype DX (21 genes)** — **both**. The Recurrence Score is an independent prognostic factor (HR 2.81 per 50-point increment of RS, $P < 0.001$) *and* it is predictive of chemotherapy benefit.

The exam-safe sentence:

> *"MammaPrint is a prognostic signature: it stratifies patients by risk of distant recurrence. Oncotype DX is prognostic and predictive: the same score is independently prognostic of recurrence and, in the randomised trial, identifies the subgroup that gains from adjuvant chemotherapy. A prognostic assay alone cannot justify withholding a treatment."*

Trap 4 — Analytical validation, clinical validity, clinical utility ⚑

These are three different things with three different proofs, and the lecture devotes a whole slide framework to them.

Stage	The question it answers	What it needs	Who does it
Analytical validation	Does the assay measure the target correctly and reproducibly?	sensitivity (LLoQ), specificity, dynamic range, linearity, precision, reproducibility (inter-day, lot-to-lot), matrix effect	the laboratory
Clinical validity	Is the measurement associated with the clinical endpoint?	a statistical correlation with the endpoint of interest	clinical / biostatistics
Clinical utility	Does using the test improve patient management and outcome?	a demonstration of benefit versus management without the test	a clinical trial

> *"Analytical precision is only the foundation; tests must ultimately demonstrate clinical validity (correlation with clinical endpoints) and clinical utility (improving patient management outcomes)."*

The exam-safe sentence:

> *"Analytical validation shows the assay measures its target accurately and reproducibly; clinical validity shows the measurement correlates with a clinical endpoint; clinical utility shows that acting on the test improves patient outcomes. An analytically perfect assay with no clinical utility changes nothing for a patient."*

Figure L7b — The biomarker and the companion diagnostic: two different things

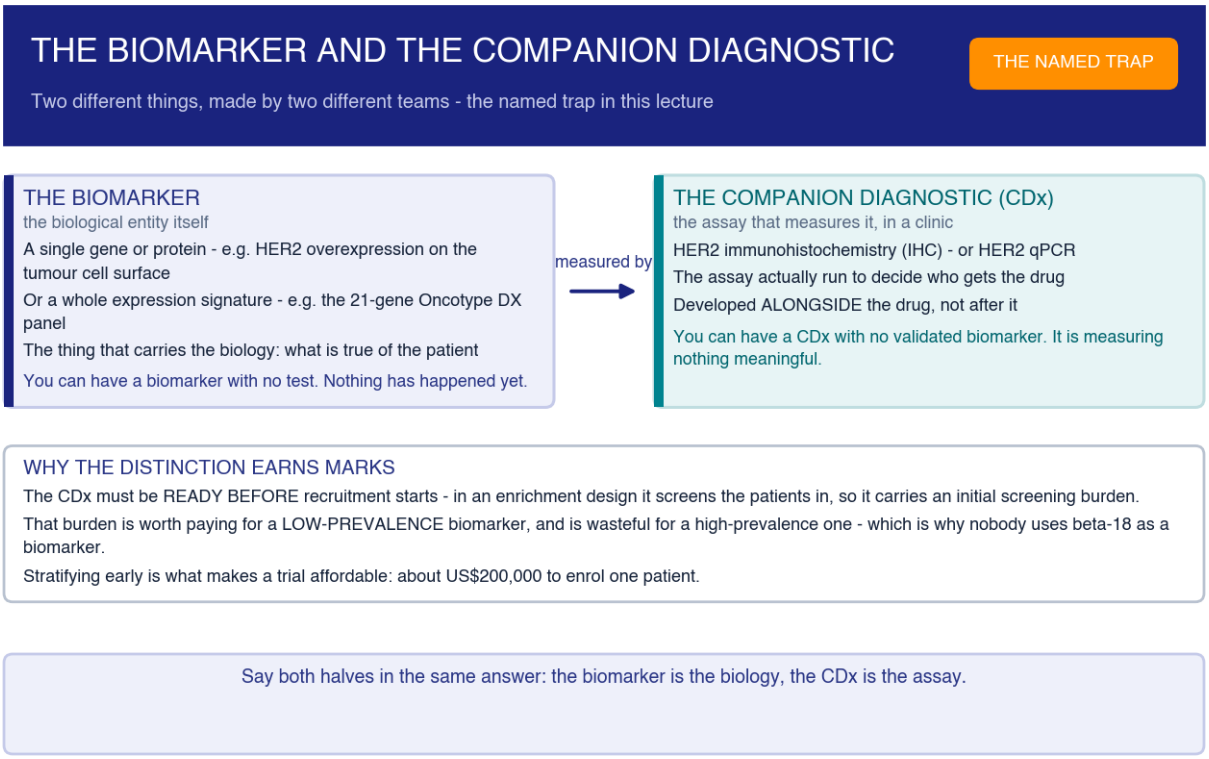


Figure L7b. The two-step concept: the biomarker is the biological entity (HER2 overexpression, or the 21-gene signature), and the companion diagnostic is the assay that measures it (HER2 IHC or HER2 qPCR), developed alongside the drug. Figure authored at 6.5 in print width and placed 1:1 on this page.

How to read this plate: the two columns are never merged. Say both halves in the same answer - the biomarker is the biology, the CDx is the assay - and you cannot be caught out by the trap. The band underneath is the consequence: the CDx must be ready before recruitment, because in an enrichment design it screens the patients in.

A3 The numbers and thresholds you must be able to recall

A3.1 Development and trial economics

Number	What it refers to
US$200,000	the cost of enrolling **one patient** in a clinical trial — the number he used to justify biomarkers: stratify early, stop a failing trial early. *(His own figure; the published median is about US$41,000 per patient — see C.2 #10.)*
US$1 million	the cost of submitting an **IND** — which is why you hold a **pre-IND meeting** first. *(His own figure; FDA's own review-cost standard for an IND is ≈US$0.46–0.62 million — see C.2 #10.)*
20–80	participants in **Phase 1**
100–300	participants in **Phase 2**
1,000–3,000	participants in **Phase 3**
1,000+	participants in **Phase 4**
Phase 1 / 2 / 3	safety & dose → preliminary efficacy → confirmatory comparative efficacy
NMPA / FDA	the Chinese (NMPA) and US (FDA) regulators to which the IND dossier is submitted

A3.2 Quantifying a rare mutation: qPCR vs ddPCR

Number	What it refers to
10 / 5 / 2.5 / 1.25 ng	the four standard-curve points for **absolute quantification by real-time PCR** — a two-fold serial dilution
4 tubes	the number of standard tubes in that example
no standard curve	the defining advantage of **droplet digital PCR**
thousands of droplets	the partitions ddPCR creates; positive/negative droplets are counted and converted by **Poisson statistics**

A3.3 The two breast-cancer signatures

	MammaPrint	**Oncotype DX**
Number of genes	**70**	**21** = **16** cancer-related + **5** reference
Platform	**DNA microarray** (70-gene chip)	**quantitative real-time PCR**
Output	**high risk / low risk** of distant recurrence	**Recurrence Score (RS), 0–100**
Risk bands	2 bands	**< 18 = low** · **18 to < 31 = intermediate** · **31 to 100 = high**
Independent prognostic factor?	—	**yes: HR 2.81 (95 % CI 1.70–4.64), P < 0.001** per **50-point** increment of RS
Sample types	fresh-frozen **and** FFPE	fresh-frozen **and** FFPE
Target patients	**Stage I or II** breast cancer; any ER/PR/HER2 status; tumour ≤ 5 cm; **node-negative**; **ER+ / HER2**−	**Stage I or II**; invasive carcinoma; tumour ≤ 5 cm; node-negative; ER+ / HER2−
Derivation	retrospective cohorts totalling 447 patients, **including the tamoxifen arm of NSABP B-20**	NSABP B-20 (plus two other cohorts)
Prospective validation	**archival blocks from NSABP B-14 — 668 of 675 analysable**	NSABP B-14
Independent validation of the 70-gene signature	**independent cohort, n = 295**	van de Vijver et al., *N Engl J Med* 2002

The RS formula, as he wrote it on the slide:

```
RS = Σ (coefficient_gene1 × expression_gene1)
+ Σ (coefficient_gene2 × expression_gene2)
+ … …
+ Σ (coefficient_gene16 × expression_gene16)
```

Coefficient = *the weighted contribution of that gene to tumour recurrence*. Range **0 to 100**.

A3.4 The chemotherapy-benefit trial (NSABP B-20)

Number	What it refers to
651	tumour tissues from node-negative, ER+ patients
227	treated with **tamoxifen alone**
424	treated with **tamoxifen plus chemotherapy**
FFPE blocks	retrospective formalin-fixed paraffin-embedded tissue blocks
HR > 1	in **low-RS** patients the odds ratio for chemotherapy exceeds 1 — no benefit, and possible harm

A3.5 The multivariate analysis (slide "RS stratifies patients by their 10-year distant recurrence")

Factor	P value	Hazard ratio (95 % CI)
Age at surgery	0.22	0.76 (0.50–1.18)
Clinical tumour size	0.38	1.19 (0.81–1.76)
Tumour grade — moderately differentiated	0.15	1.55 (0.85–2.81)
Tumour grade — poorly differentiated	**< 0.001**	**3.34 (1.79–6.26)**
HER2 amplification	0.06	0.51 (0.26–1.02)
ER protein 50–99 fmol/mg	0.32	0.75 (0.43–1.31)
ER protein 100–199 fmol/mg	0.72	0.90 (0.52–1.58)
ER protein ≥ 200 fmol/mg	0.94	1.02 (0.58–1.70)
Recurrence score	**< 0.001**	**2.81 (1.70–4.64)**

Read this table for what it is: apart from **poorly differentiated grade**, the Recurrence Score is the only factor whose confidence interval excludes 1. Age, tumour size, HER2 amplification and ER level are all non-significant. That is exactly why it can be used clinically.

Two cautions. First, the hazard ratio **2.81 (1.70–4.64) is per 50-point increment of the score**, not per single point — saying "per unit" overstates it fifty-fold (see C.2 #1). Second, the individual covariate hazard ratios in this table could not be checked against an accessible copy of the paper, so quote them as *the slide's numbers* and verify against Paik et al. 2004 before using them as sourced values (C.5).

A3.6 The microRNA classifier for early HCC

Number	What it refers to
n = 96	HCC tumour + adjacent non-tumour tissues profiled in the discovery step
6 → 4 → 2	miRNAs selected at each narrowing step, ending in **miR-15b + miR-130b**
n = 15	HCC patients whose serum was taken **before and after surgical removal** of the tumour (longitudinal)
n = 30 / 29 / 57	independent validation cohort: healthy controls / chronic hepatitis B carriers / HCC patients (**116** in total)
AUC = 0.981	area under the ROC curve for the miR-15b + miR-130b classifier
98.25 %	sensitivity
91.53 %	specificity
94.83 %	accuracy
2 cm	the distance from the tumour edge at which **cancer-adjacent tissue** was sampled

A3.7 Analytical validation numbers

Number	What it refers to
10–100 ng/mL	the **dynamic range** of an assay that fails clinically
100–1,000 ng/mL	the **pathological range** in cancer patients — the two do not overlap, so the assay is useless
CDH17 vs CDH16 vs CDH1	the specificity problem: Cadherin-17 must be distinguished from its homologues Cadherin-16 and Cadherin-1
2024 vs 2026	antibody lots produced in different years by a CDMO — they must give the same signal, or the assay is not reproducible
3 matrices	the biological matrices used in the matrix-effect spike-in experiment
LLoQ	lower limit of quantification — the floor of analytical sensitivity

Figure L7p — Prognostic against predictive, and what the Recurrence Score decides

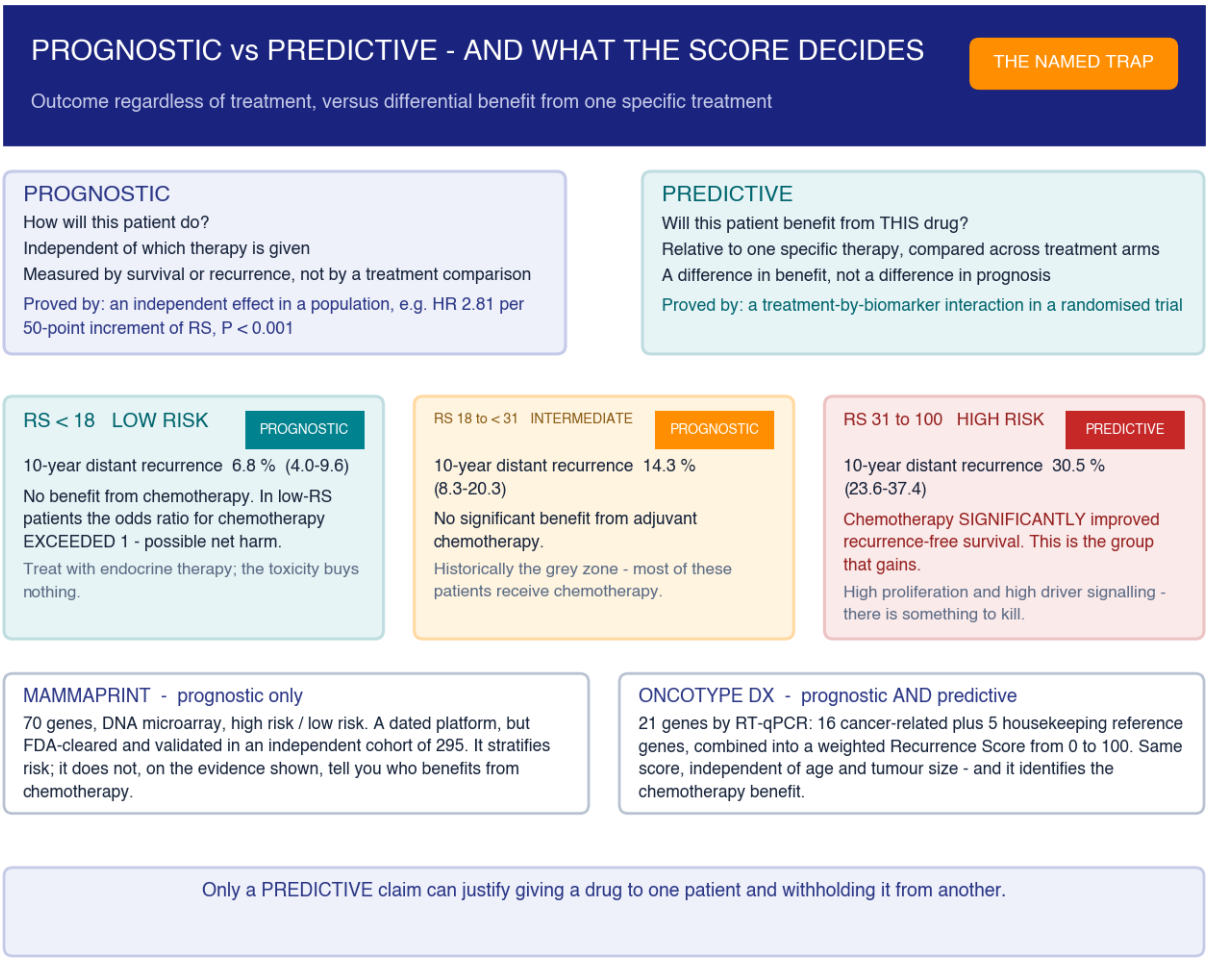


Figure L7p. The two words, the three Recurrence Score bands with their 10-year distant recurrence rates and their chemotherapy benefit, and the two assays that produce them. Figure authored at 6.5 in print width and placed 1:1 on this page.

How to read this plate: the top row is the definition and the bottom row is the evidence. The single sentence to memorise is in the footer - only a predictive claim can justify giving a drug to one patient and withholding it from another.

A4 The discriminator pairs you must be able to separate

Each of these is a pair he put side by side on a slide, or stated as a contrast in words. If a question gives you an option for each half, the pair is the answer.

No.	The pair	The line that separates them
1	**Driver vs passenger**	a driver **causes** the oncogenic transformation; a passenger is co-expressed or co-mutated and does nothing
2	**Biomarker vs companion diagnostic (CDx)**	the biomarker is the biology; the CDx is the assay that measures it
3	**Prognostic vs predictive**	prognosis is about outcome; prediction is about differential benefit from a specific treatment
4	**Analytic vs clinical validation**	analytic = does it measure right; clinical = is it associated with the endpoint
5	**Clinical validity vs clinical utility**	validity = correlated with the endpoint; utility = using it changes management and improves outcome
6	**qPCR vs ddPCR**	qPCR needs an external **standard curve**; ddPCR needs none and counts positive/negative droplets by **Poisson** statistics
7	**Real-time PCR vs DNA microarray**	PCR measures **specific transcripts by sequence**; microarray measures **what is on the chip** by hybridisation
8	**MammaPrint vs Oncotype DX**	70 genes on a **microarray**, 2 risk bands, prognostic; 21 genes on **RT-qPCR**, 3 risk bands, prognostic **and** predictive
9	**Tumour tissue vs tumour-adjacent tissue**	both are prognostic; the adjacent tissue's "inactive" signature is enriched for **anti-tumour immunity**
10	**"Active" vs "inactive" adjacent signature**	active = worse survival; inactive = better survival, anti-tumour immunity genes
11	**Small molecule vs antibody**	a small molecule needs **target specificity**; an antibody needs a **well-tuned affinity** — too high and its blood half-life suffers
12	**Enrichment design vs all-comer design**	enrichment screens by the biomarker before enrolment (needs a ready CDx, good for low-prevalence markers); all-comer enrols everyone and correlates afterwards
13	**Autoantibody vs anti-drug antibody (ADA)**	an autoantibody **amplifies** a real signal and makes a good biomarker; an ADA **interferes** with the assay and can neutralise the drug
14	**PD marker vs PK marker**	PD = the drug's **effect** on the target; PK = the drug's **level in the body** over time
15	**Direct vs indirect PD marker**	a direct marker is the target itself (HER2 for trastuzumab); an indirect marker is a downstream readout (a gene activated downstream of the target)
16	**Affinity vs specificity** (antibody vs small molecule)	affinity = how tightly it binds; specificity = how little else it binds
17	**Screening vs confirming mutation**	a **panel** of variants is screened in clinic; one mutation (e.g. T790M) is what drives a decision
18	**Intrinsic vs extrinsic matrix effect**	intrinsic = the specimen's own proteins; extrinsic = the container or the assay reagents

A5 Method selection: "given this question, which technique?"

The question you are asked	The method	Why	What it still cannot do
Which pathways are up- or down-regulated between two groups?	**Pathway enrichment analysis** (g:Profiler, GSEA) + **Cytoscape/EnrichmentMap** for the network	turns a DEG list into biology; genes act in pathways, not alone	it cannot tell a driver from a passenger on its own — that needs CNV as well
Is this gene a driver in this tumour?	**CNV analysis + PEA together**	a driver shows **both** up-regulation **and** copy-number change, in an enriched pathway	it is an inference, not a proof
Is this tumour driven by EGFR? Is the patient resistant?	**Real-time PCR (TaqMan)**, or **ddPCR** from plasma	detects the specific resistance mutation such as **T790M**	a single assay covers one mutation; a **panel** covers many
Can I quantify a rare mutant without a standard curve, in a tiny volume, several targets at once?	**Droplet digital PCR**	absolute counting by Poisson statistics; **multiplex**; ideal for low-abundance **ctDNA**	still needs validated primers and a proper extraction
What is this patient's risk of distant recurrence, and does she need chemotherapy?	**Oncotype DX (21-gene RT-qPCR)** → **Recurrence Score**	combines prognosis and a treatment-by-biomarker interaction from a randomised trial	it is **not** for HER2+ or late-stage disease
What is this patient's risk of distant recurrence (early stage, any receptor status)?	**MammaPrint (70-gene microarray)**	the 70-gene signature with independent-cohort validation	prognostic only; microarray is a dated platform
Can I detect early liver cancer from a blood sample?	**Circulating miRNA panel** (miR-15b + miR-130b) + **ROC curve**	AUC 0.981 with high sensitivity and specificity; longitudinal sampling shows the level falls after tumour removal	needs validation in a real screening population
Is this assay ready to run in a clinic?	**Analytical validation**	LLoQ, specificity, dynamic range, reproducibility, matrix effect	a perfect assay can still have no clinical utility
Should we use this test to change a patient's treatment?	**Clinical utility** (trial evidence)	the only stage that demonstrates an outcome benefit	—
Can I find the patients who will respond to this drug?	**Enrichment design** with a validated **CDx**	screens by biomarker before enrolment; shorter, cheaper trials	needs the CDx ready *before* the trial; wasteful for low-prevalence markers
Can this drug perturb a gene in a human-relevant system?	**CRISPR/Cas9** knock-out/knock-in → **patient-derived organoids** / **liver-chip**	isogenic controls and human physiology, not just cell lines	an organoid is still a model, not the patient

Figure L7v — From a measurement to a clinic: three rungs, three different proofs



Figure L7v. Analytical validation, clinical validity and clinical utility, each with the question it answers and the evidence it needs; the five analytical parameters with the failure each one catches; and the two antibodies with opposite effects. Figure authored at 6.5 in print width and placed 1:1 on this page.

How to read this plate: climb the rungs from the bottom. The left column is the ladder he flagged as the likely exam focus; the right column is what makes each rung hard. The last line of the right column is the one that catches people: an autoantibody amplifies a signal you want, and an anti-drug antibody corrupts the assay you are trying to run.

A6 The English answer skeleton, and the habits that earn marks

A6.1 The four moves

Every answer he will accept is built out of the same four moves. Use them in this order.

1	**Name the thing.** One line: what this molecule, gene, assay or score *is*.
2	**Give the mechanism.** One to three lines: what it does, and in what direction.
3	**Give the consequence.** One line: what that means for the patient, the experiment or the trial.
4	**State the limitation.** One line: what it still cannot do, or what would confound it.

The fourth move is the one students skip, and it is the one that turns a correct answer into a complete one. Dr. Wong said in Lecture 6 that describing data when the question said *explain* was the most common midterm mistake — and the same habit will cost you marks here.

A6.2 Marker-pleasing habits

- **Lead with the discriminator.** If the question offers "prognostic or predictive", your first sentence must contain the word. Do not build up to it.
- **Give the number.** "The Recurrence Score is 0 to 100" is worth more than "the RS is a score".
- **Use the arrow chain.** Write the mechanism as `A → B → C`, not as prose. The arrow is what marks.
- **Name the tool.** If he asked which software, say **g:Profiler**, **GSEA**, **Cytoscape**, **EnrichmentMap** — the tool names are the content.
- **State the assay, not the analyte, when asked for the CDx.** "HER2 IHC", not "HER2".
- **Say what the cut-off is.** RS < 18 / 18 to < 31 / 31 to 100.
- **One English sentence per idea.** Short and clear, as he told you.

A6.3 Two templates

Template 1 — "Explain how X works."

> *"X is [what it is]. It acts by [mechanism], so that [effect]. As a result, [consequence for the patient / the trial]. It cannot, however, [limitation]."*

Template 2 — "Which method would you use, and why?"

> *"I would use [method] because [reason tied to the specific feature of the question]. It is preferred over [alternative] because [the decisive difference]. The method's limitation is [limitation], which in this study is handled by [what they did about it]."*

A7 What NOT to memorise

He released these explicitly; do not spend revision time on them.

1	**The gene IDs of the 21-gene panel.** *"don't memorize all this genes. ID is not necessary."* Know the five functional groups instead.
2	**The full list of genes in the 70-gene MammaPrint signature.** Know it is 70 genes and that it came from ~25,000 genes on a microarray.
3	**The individual CDH17 / CDH16 / CDH1 sequences.** Know that homology is why antibody specificity must be tested, and that you screen against CDH16 and CDH1.

4	**The detailed contents of the FDA pharmacogenomic biomarker table.** Know that **CYP2C19** is the biomarker for **abrocitinib (NDA 213871)** and that the label halves the starting dose to **50 mg once daily** for a poor metaboliser, rising to 100 mg after 12 weeks. *(The deck prints the wrong NDA number — see C.2 #8.)*
5	**The minute-by-minute order of the drug-development pipeline.** Know the *sequence*: target → hit → lead → in vitro/in vivo → toxicology → IND → Phase 1 → 2 → 3 → approval.
6	**Every figure in the deck.** *"Don't be scared by all these figures."* You are not being examined on the artwork.

A8 Self-test — 15 questions, closed book, English

Sit these properly: closed book, English, short answers. Mark yourself against the key only when you have finished all fifteen.

Q1	Explain, in one English sentence each, the difference between a biomarker and a companion diagnostic, using HER2 as the example.
Q2	A patient with node-negative, ER-positive breast cancer has an Oncotype DX Recurrence Score of

24	Is the score prognostic, predictive, or both? Justify each answer in one further sentence.

Q3	A trial enrols every patient with advanced solid tumours regardless of any biomarker, measures the biomarker retrospectively, and reports that biomarker-positive patients did better on drug A while biomarker-negative patients did better on drug B. Name the trial design and state what the biomarker is.
Q4	Explain why a tumour must be profiled alongside the histologically normal tissue 2 cm from its edge, and what an "inactive" signature in that adjacent tissue indicates.
Q5	Give the four standard-curve concentrations used for absolute quantification of a mutant allele by real-time PCR, and state the defining advantage of droplet digital PCR over that method.
Q6	A new ELISA for a plasma protein measures reliably from 10 to 100 ng/mL. In cancer patients the protein circulates at 100 to 1,000 ng/mL. Name the analytical failure and state what must be changed.
Q7	You are developing an antibody against Cadherin-17. Explain, with reference to CDH16 and CDH1, why your antibody must be screened against them.
Q8	State the Recurrence Score risk bands, and state in which band(s) adjuvant chemotherapy significantly improved recurrence-free survival in NSABP B-20, and in which band it did not.
Q9	A tumour shows copy-number amplification and high expression of a gene that sits in a strongly enriched pathway. Is that gene a driver or a passenger? State the two pieces of evidence you would need to separate them and explain why one alone is not enough.
Q10	Explain why holding a pre-IND meeting before submitting an IND application is worth doing, and give the figure Dr. Wong gave for the cost of an IND submission.
Q11	Distinguish pharmacokinetics from pharmacodynamics, and give the HER2/trastuzumab example of a direct pharmacodynamic marker.
Q12	What is the difference between an autoantibody and an anti-drug antibody, in terms of their effect on a therapeutic antibody assay?
Q13	Name the four software tools shown in the pathway enrichment workflow and state the role of each.
Q14	Your collaborator reports that a gene signature discovered in a retrospective cohort of 300 patients "validates" the signature because it separates survival curves in that same cohort. Explain the error and state the requirement.
Q15	A tumour-adjacent tissue signature predicts recurrence. Explain how this changes what a surgeon should do at the operation, and what it implies about the tissue margin.

Answer key

Q1	A biomarker is the biological entity itself — here, HER2 overexpression on the tumour cell surface. The companion diagnostic is the assay that measures it in the clinic — here, HER2 immunohistochemistry (or HER2 qPCR). The biomarker is what you are measuring; the CDx is how you measure it, and the CDx is what actually gets run to select patients for the trial.
Q2	**Both.** *Prognostic*: the Recurrence Score is an independent predictor of distant recurrence in multivariate analysis (HR 2.81 per 50-point increment of RS, 95 % CI 1.70–4.64, P < 0.001), independent of age and tumour size. *Predictive*: in the randomised trial, the score separated the patients who gained from adjuvant chemotherapy from those who did not — benefit was confined to the high-RS group, and in low-RS patients the odds ratio for chemotherapy exceeded 1. Prediction requires a treatment arm; the prognostic claim does not.

Q3	The **all-comer design**. The biomarker is **predictive** — a biomarker that is only prognostic would not produce a differential treatment effect. In an all-comer design the biomarker is measured retrospectively and its value is inferred from the treatment-by-biomarker interaction.
Q4	Because the tissue next to the tumour carries prognostic information of its own, independent of the tumour itself. An "inactive" adjacent signature is associated with **better** survival, and gene ontology showed that the genes in it are enriched for **anti-tumour immunity** — i.e. it marks an active local immune response that suppresses micrometastases.
Q5	**10 ng, 5 ng, 2.5 ng, 1.25 ng** — a two-fold serial dilution across four tubes. ddPCR's defining advantage is that it needs **no standard curve**: it partitions the sample into thousands of droplets, amplifies each, and counts positive versus negative droplets, converting the counts to absolute numbers by **Poisson statistics**. It also multiplexes and suits low-abundance ctDNA.
Q6	A **dynamic-range failure**: the assay's linear range (10–100 ng/mL) does not overlap the clinical pathological range (100–1,000 ng/mL), so the assay cannot read the values that matter. The assay must be re-developed (or the measurement range extended) so that its quantification range covers the clinical range.
Q7	Cadherin-17 shares substantial sequence homology with **Cadherin-16 (CDH16)** and **Cadherin-1 (CDH1)**. An antibody raised against CDH17 may therefore bind CDH16 or CDH1, which would give a false positive signal. Every candidate clone must be counter-screened against CDH16 and CDH1, and only clones with strict CDH17 specificity are taken forward. Nucleic-acid assays do not have this problem to the same degree, because a probe's specificity comes from base-pairing with its own sequence.
Q8	**< 18 = low risk; 18 to < 31 = intermediate; 31 to 100 = high.** Chemotherapy significantly improved recurrence-free survival in the **high-RS** group only. In the **low** and **intermediate** groups there was no significant benefit, and in low-RS patients the odds ratio for chemotherapy exceeded 1, which suggests possible harm from the toxicity without the anti-tumour benefit.
Q9	It is more likely a **driver**, because it has copy-number amplification *and* up-regulation *and* sits in an enriched pathway. But neither observation alone is sufficient: up-regulation without a copy number change may be a passenger driven by the tumour's overall transcriptional state, and copy-number change without enrichment may be a passenger in a bystander pathway. The convergence of **expression + CNV + pathway enrichment** is what separates driver from passenger.
Q10	Because an IND submission is expensive — Dr. Wong's figure is **US$1 million** — and because an unanswered question from the regulator generates more issues and delays. The pre-IND meeting is where you submit your data and ask whether it is sufficient, and what further experiments the agency requires, before you spend the money. He assigns this job to the regulatory affairs specialist: *"you need to ask the right question, at the right time, the right moment."*
Q11	**Pharmacokinetics** is the level of the drug in the body over time after administration (absorption, distribution, metabolism, excretion — ADME). **Pharmacodynamics** is the effect of the drug molecule on its target. **HER2** is a *direct* pharmacodynamic marker for **trastuzumab (Herceptin)**, because the drug's target is the HER2 receptor and HER2 status is what the drug acts on. A direct PD marker is the target itself; an indirect PD marker is a downstream readout — for example, the expression of a gene activated downstream of the target, used when the target itself is hard to measure.
Q12	An **autoantibody** is the patient's own immune response to a tumour neoantigen. Because immune signal amplification makes the antibody titre far higher than the circulating neoantigen concentration, autoantibodies are **highly sensitive diagnostic targets** — they amplify the signal you want. An **anti-drug antibody (ADA)** is mounted against the administered therapeutic antibody, which the immune system sees as foreign. ADAs are an **interference**: they compete with the assay's capture and detection antibodies (corrupting pharmacokinetic quantification) and can **neutralise** the therapeutic antibody.
Q13	**g:Profiler** — a web server for pathway and GO-term enrichment of a gene list, offering multiple enrichment methods. **GSEA (Gene Set Enrichment Analysis)** — finds statistically enriched gene sets from an expression-ranked list, rather than from a cut-off list of genes. **Cytoscape** — builds and views the interaction network of the enriched pathways and their shared genes. **EnrichmentMap** — a Cytoscape app that visualises the enrichment results as a network, so you can see which pathways overlap and which are connected.
Q14	The error is **validating a signature in the cohort it was discovered in.** The requirement is **an independent cohort** — *"you cannot validate your finding in a single cohort, you always need to validate your finding in an independent cohort."* Discovery in retrospective samples from a research laboratory establishes the hypothesis; only an independent, separately recruited cohort establishes that the signature holds in a different population. This is exactly how MammaPrint's 70-gene signature was validated in a separate cohort of **295** patients.
Q15	It changes the operation from *"cut out this lump"* to *"treat this field."* If the adjacent tissue carries an adverse ("active") signature — meaning the microenvironment around the tumour is biologically hostile — then a narrow margin leaves hostile tissue behind and the recurrence risk persists. The margin has to be planned with the biology of the adjacent tissue in mind, and a patient with a favourable ("inactive", immunity-enriched) adjacent signature is a lower-risk patient even though their tumour looks identical.

PART B — LEARN FROM ZERO

Ten concept chains that take you from zero clinical genomics to being able to explain the lecture rather than recognise it. Work through them in order, with a pen in hand.

B0 What you need before you start

You need four things from earlier lectures. If any of them is shaky, go back to that lecture first — this one stacks on all four.

From	What you must already know
Lecture 1–2	what a gene, a protein and a pathway are; how the central dogma works; what "expression" means
Lecture 3	that DNA is packaged into chromatin, so "switching genes on and off" is a structural event; what a promoter and an enhancer do
Lecture 4	how DNA is replicated and repaired; what **NHEJ** and **HR** are, because CRISPR uses both
Lecture 5	how gene expression is controlled; what a transcription factor does
Lecture 6	the techniques: real-time PCR and Ct, microarray vs RNA-Seq, western blot, ELISA, mass spectrometry, CRISPR screens, transfection

You also need one idea from this course's earlier material that Dr. Wong keeps reusing: **a cancer is not only its tumour cells**. The tumour sits in tissue, in a microenvironment, with an immune infiltrate and a blood supply. When you profile a tumour, you are profiling a mixture.

B1 The ten concept chains

A chain is a line of reasoning. Each one ends with a single comparison you must be able to say out loud, because that comparison is what the exam asks for.

Chain 1 — From a table of genes to a statement about biology

The problem. After an RNA-Seq or microarray experiment you have a list of a few hundred differentially expressed genes. That list is not an answer. It is a pile of names, and the number of genes alone tells you nothing about the biology.

The move. Put the list through a **pathway enrichment analysis (PEA)**. The tool asks: compared with a random set of the same size from the same genome, are these genes over-represented in any known pathway? If the Ras signalling pathway is enriched, the experiment is not "300 genes changed"; it is "the Ras pathway is deregulated in this tumour."

The tools on the slide. g:Profiler takes your gene list and returns enriched pathways and GO terms, with several enrichment methods available. **GSEA** takes a different input — a *ranked* list of all genes by how much they changed — and asks which gene sets are concentrated at the top or bottom of that ranking; it therefore does not depend on an arbitrary fold-change cut-off. **Cytoscape** draws the result as a network, because pathways overlap and share genes. **EnrichmentMap** is the Cytoscape app that turns a table of enriched terms into that network, so you can see which pathways are connected.

Why the network matters. Genes do not act one at a time. A network view is how you see that one enriched pathway feeds another. Dr. Wong returned to his Lecture 1 cartoon here: a bioinformatician who does not annotate their data properly produces a list a co-worker cannot interpret.

The comparison you must say out loud:

> *"Pathway enrichment turns a list of genes into a statement about which biological process is deregulated — and because the network view shows pathways feeding each other, it also tells you the process is not isolated."*

Chain 2 — Driver or passenger?

The problem. In a tumour, some gene changes cause the cancer and some are just along for the ride. A tumour has hundreds of mutations and the expression of thousands of genes is disturbed. If you only look at expression, you cannot tell which changes matter.

The move. Bring in a second, independent kind of evidence: **copy number variation (CNV)**. A driver gene is often amplified (or deleted) *and* up-regulated *and* sits inside a pathway that is enriched. A passenger gene is typically none of those — it may be up-regulated purely because the tumour's overall transcriptional state has changed, or amplified by chance.

Why one piece of evidence is not enough. Expression alone is confounded: a passenger in a highly active pathway will be up-regulated. CNV alone is confounded: a bystander region can be amplified without any consequence. Only the **convergence** of expression, CNV and pathway enrichment points at a driver.

The comparison you must say out loud:

> *"A driver gene drives the oncogenic transformation; a passenger is altered but non-causal. Expression alone cannot separate them, because passengers are co-expressed with drivers — you need copy-number evidence and pathway enrichment as well."*

Chain 3 — The biomarker and the CDx: two different things

The problem. In a paper you will read "HER2-positive patients benefited from trastuzumab". It sounds like one fact. It is two, and they are made by different people in different places.

The move. Split them.

- The **biomarker** is the biological entity: HER2 overexpression on the tumour cell surface.
- The **companion diagnostic (CDx)** is the assay: HER2 immunohistochemistry, or HER2 qPCR, run in a laboratory to decide whether this patient goes on the drug.

The CDx is developed **alongside** the drug, not after it. That is why the assay has to be *ready* before the trial starts: in an **enrichment design**, you screen every incoming patient with the CDx and only enrol those who test positive.

Why stratification is worth the trouble. Dr. Wong's economics: **US$200,000 per patient enrolled**. Enrol only the patients likely to respond and the trial gets smaller, faster and more likely to succeed. It is also the ethical case: it avoids giving a toxic drug to patients who cannot benefit.

The one practical trap. An enrichment design has an **initial screening burden**, and it is a good design for **low-prevalence** biomarkers. If your biomarker is expressed in only a small fraction of patients, screening everyone to find them is wasteful — which is exactly why he noted that nobody uses **β18** as a biomarker for that reason.

The comparison you must say out loud:

> *"The biomarker is the biology; the companion diagnostic is the assay that measures it. The CDx is developed alongside the drug and must be ready before recruitment starts, because in an enrichment design it screens the patients in."*

Chain 4 — Finding a resistance mutation in a drop of blood

The problem. A patient on a tyrosine kinase inhibitor develops resistance. The tumour has acquired a secondary point mutation in the kinase domain — for example **EGFR T790M** — that changes the shape of the ATP-binding pocket so the inhibitor no longer fits. Continuing the drug is now futile, and you need to know quickly.

Why you need a liquid biopsy. Re-biopsying a patient is invasive and slow. Tumour cells die and shed their DNA into the blood as **circulating tumour DNA (ctDNA)**, a small fraction of the total **cell-free DNA (cfDNA)**. A blood sample is fast and repeatable.

The move, and the two technologies.

- **Real-time PCR (qPCR)** measures how much target is present by the cycle at which fluorescence crosses a threshold (the **Ct**). To get an **absolute** number you must build an **external standard curve**: amplify known amounts of the target — in his example, a two-fold series of **10, 5, 2.5, 1.25 ng** — and read the unknown sample's Ct off that curve. Lower starting template, higher Ct.
- **Droplet digital PCR (ddPCR)** does something different in kind. It partitions the sample into **thousands of droplets** in a microfluidic emulsion, amplifies the target in each, and then simply **counts the fluorescent-positive and negative droplets**. The ratio of positives to negatives gives the absolute concentration by **Poisson statistics** — **no standard curve at all**.

Why ddPCR wins for this job. Three reasons, all from the lecture: **absolute quantification without a standard curve**; **sensitivity** for the very low abundance of ctDNA; and **multiplexing** — several targets in one tube from a minimal sample, which matters when the sample is a drop of blood and the reaction-to-reaction variation must be minimised.

Where the biology comes from. Dr. Wong's own industry work at **Johnson & Johnson**: he cultured lung cancer cell lines under escalating TKI concentrations and then whole-genome sequenced the survivors. The cells had acquired **the same resistance mutations** seen in treated patients. That is the argument for the whole pipeline: in vitro selection predicts clinical resistance, which is how you design the second- and third-generation inhibitors *before* patients need them.

The comparison you must say out loud:

> *"Real-time PCR needs an external standard curve to give an absolute number; droplet digital PCR needs none — it partitions the sample, counts positive and negative droplets and converts that by Poisson statistics — which is why it suits low-abundance ctDNA and multiplexing."*

Chain 5 — The 70-gene signature, and the rule that makes it real

The problem. You want to tell a patient with early breast cancer whether she is likely to relapse. You could measure one marker, but cancer is not a single-gene disease.

The move. Use a **multi-gene expression signature**. MammaPrint's 70 genes were derived from ~25,000 genes measured on a **DNA microarray**, narrowed from 231 regulated genes down to 70. The genes cover the hallmarks of cancer — invasion, metastasis, angiogenesis.

The rule that makes the signature real. A signature discovered in a retrospective sample set from a research laboratory is a **hypothesis**. It becomes a clinical test only after it is validated in an **independent cohort** — different patients, different period, separately recruited. The slide shows the validation: the 70-gene signature split **295** independent patients into two survival curves, a poor overall survival group and a better one, and the same for metastasis-free survival.

Why the assay is still a microarray. Dr. Wong is blunt: microarray is *"quite obsolete technology"* for research — RNA-Seq is cheaper, more effective, and finds novel transcripts and splice isoforms that a fixed probe set cannot. But MammaPrint is **FDA-cleared**, standardised and in clinical use, so it remains clinically legitimate. A dated platform is not a wrong platform if it is validated and approved for the job.

Where it applies. Stage I or II breast cancer — early disease. Not late stage: an advanced patient already has a high baseline risk of distant recurrence, so a prognostic score tells you nothing you did not already know. Any ER / PR / HER2 status is eligible. Both **fresh-frozen** and **FFPE** tissue work.

The comparison you must say out loud:

> *"A microarray is a dated platform but MammaPrint is a validated, FDA-cleared 70-gene prognostic signature; and a signature discovered in one retrospective cohort has no clinical value until it has been validated in an independent cohort."*

Chain 6 — The 21-gene score and how it is built

The move. Oncotype DX does the same job with far fewer genes and a different platform: **21 genes by RT-qPCR**, of which **16** are cancer-related and **5** are reference (housekeeping) genes.

Why the reference genes are there. Housekeeping genes such as **GAPDH** and **ACTB (β-actin)** are expressed at a stable level across cells and samples. They are the internal control: they tell you the extraction and the RT worked, and they normalise the sample-to-sample variation so that the cancer-gene measurements are comparable between patients.

Why 21 and not 70. RT-qPCR measures specific transcripts by sequence, needs no array, no hybridisation and no preset probe set, and runs from FFPE tissue. That is a far more practical assay for a routine clinical laboratory.

The scoring. Each gene's expression level is multiplied by a **coefficient** — its weighted contribution to recurrence. *HER2* carries a higher coefficient than *ST3* because it is a key driver; *ST3* is MMP-11, a matrix metalloproteinase involved in invasion. The weighted sum is the **Recurrence Score**, which runs **0 to 100** and is banded:

```
RS < 18 → low risk

RS 18 to < 31 → intermediate risk

RS 31 to 100 → high risk
```

Where it applies. Early breast cancer: **Stage I or II**, invasive carcinoma, tumour **≤ 5 cm**, **node-negative**, **ER+ / HER2−**; FFPE or fresh specimens.

The comparison you must say out loud:

> *"Oncotype DX is 21 genes by RT-qPCR — 16 cancer-related plus 5 housekeeping reference genes that normalise the assay — combined into a weighted Recurrence Score from 0 to 100, banded at < 18, 18 to < 31 and 31 to 100."*

Chain 7 — From prognosis to a treatment decision

This is the heart of the lecture. Chain 6 gave you a number. Chain 7 asks what you may *do* with it.

First, is it prognostic? In the multivariate analysis, the Recurrence Score was the only factor with a hazard ratio whose confidence interval excluded 1 (other than poorly differentiated grade): **HR 2.81 (1.70–4.64), P < 0.001** — and note carefully, this hazard ratio is computed for a **50-point increment** in the score, not for one point. Age, tumour size, HER2 amplification and ER level were all non-significant. So the score carries information independent of the classical clinical parameters.

Then, is it predictive? That requires a **randomised trial with a treatment arm** — NSABP B-20: **651** node-negative ER+ patients, treated with **tamoxifen alone (n = 227)** or **tamoxifen plus chemotherapy (n = 424)**, on retrospective FFPE blocks.

The result, and it is the whole point of the chain:

- **High RS** — chemotherapy significantly improved recurrence-free survival. The high-risk patients are exactly the ones with the most to gain.
- **Low and intermediate RS** — no significant benefit. In low-RS patients the **odds ratio for chemotherapy exceeded 1**, meaning the toxicity may have done net harm.

The mechanism, as a chain. High RS means high proliferation and high driver signalling, so the recurrence risk on endocrine therapy alone is high, and cytotoxic chemotherapy removes the proliferating cells. Low RS means low intrinsic proliferation, so there is little to kill and the toxicity is all that is left.

The sentence that earns the mark:

> *"The Recurrence Score is prognostic because it is an independent predictor of recurrence (HR 2.81 per 50-point increment of RS, 95 % CI 1.70–4.64, P < 0.001), and predictive because in a randomised trial it identified the subgroup that gained from chemotherapy; benefit was confined to the high-RS group, and in low-RS patients the odds ratio for chemotherapy exceeded 1."*

Chain 8 — The tissue next to the tumour

The problem. Everyone profiles the tumour. But the tumour is not alone; it sits in tissue that has its own biology, its own immune infiltrate, and its own behaviour.

The study. In breast cancer, gene expression was profiled in the tissue **adjacent to the tumour** — histologically normal, non-cancerous — as well as in the tumour. A signature was found in the adjacent tissue that was strongly related to the patient's clinical outcome.

The result. Patients with an **"active"** adjacent signature had **poorer** survival; patients with an **"inactive"** signature did better. When the genes in the inactive signature were examined, they were enriched for **anti-tumour immunity** — that is, the "inactive" label describes a tissue that is quietly mounting an effective immune response against the tumour, and that response suppresses micrometastases.

Why it matters clinically. It changes what the surgeon should be doing. If the field around the tumour is hostile, cutting a narrow margin leaves hostile tissue behind and the recurrence risk stays high. The margin decision is a biological decision.

The comparison you must say out loud:

> *"Profiling the tumour alone is not enough — the histologically normal tissue adjacent to it carries its own prognostic signature, and its 'inactive' signature is enriched for anti-tumour immunity genes, so the microenvironment around the tumour matters as much as the tumour itself."*

Chain 9 — Turning a measurement into a test: the validation ladder

This is the block he flagged as the exam focus. It is a ladder, and you climb it in order.

Step 0 — the team. A biomarker is not made by one scientist. You need **discovery scientists**; **business development / commercial** people, who stop you building a redundant *"me-too"* assay in a crowded market; **medical writers**, who compile the regulatory dossier; **QA and GLP** specialists, who enforce Good Laboratory Practice — including the mandatory **non-human primate (monkey) toxicology** studies; **clinical operations** and **regulatory affairs** specialists, who run the trials and manage the **pre-IND** and **IND** meetings; and a **project manager** to chase timelines. Smaller companies bring in external consultants and subject-matter experts.

Step 1 — analytical validation. Does the assay measure its target correctly?

- **Sensitivity** — the **lower limit of quantification (LLoQ)**: the lowest amount of target the assay can reliably quantify.
- **Specificity** — does it bind only the target? Nucleic-acid assays get specificity from base pairing and are relatively easy; **protein immunoassays are not**, because homologous proteins share epitopes. His own example: **Cadherin-17 (CDH17)** is homologous to **CDH16** and **CDH1**, so every candidate antibody must be counter-screened against both. **TaqMan** real-time PCR gets specificity from probe design.
- **Dynamic range** — the assay's linear range must cover the **clinical** range. An assay reading 10–100 ng/mL is useless when patients carry 100–1,000 ng/mL.
- **Reproducibility** — the same signal across **different days**, **different operators** and **different reagent lots**. He makes the point with antibody batches made in **2024** and **2026** by a CDMO: if the 2026 lot binds differently, the assay has drifted.

Step 2 — matrix effects. The specimen is not pure analyte. **Intrinsic** interferents are the specimen's own high-abundance proteins — **albumin**, **immunoglobulins** — and **extrinsic** ones come from the container or the reagents. Two special cases:

- **Autoantibodies** are the patient's own antibodies to a tumour neoantigen. Because the immune system amplifies the signal, autoantibody titres are far higher than the circulating neoantigen itself — so they are **excellent, sensitive biomarkers**.
- **Anti-drug antibodies (ADAs)** are made against the *therapeutic* antibody. They are an **interference**: they compete with the assay's capture/detection antibodies, corrupting pharmacokinetic measurement, and can **neutralise** the drug. They are also why a sandwich ELISA needs blanks.

Step 3 — clinical validation. Two distinct claims:

- **Clinical validity** — the measurement correlates with the clinical endpoint (ROC curve, cut-off, sensitivity, specificity).
- **Clinical utility** — using the test actually improves management and outcomes.

He is explicit that a test can be analytically perfect and still useless: *"a biomarker is useless unless it has clinical utility."*

The comparison you must say out loud:

> *"Analytical validation proves the assay measures its target accurately and reproducibly — sensitivity, specificity, dynamic range, precision and matrix effect; clinical validity proves it correlates with the clinical endpoint; clinical utility proves that using it improves patient outcomes. Only the third one changes anything for a patient."*

Chain 10 — Perturbing the genome to ask a causal question

The last block is experimental: how do you find out what a gene *does*?

The move. Use **CRISPR/Cas9** to knock a gene out, or to knock something in.

- **Knock-out**: Cas9 cuts the target and the cell repairs the break by **NHEJ** (error-prone), producing indels and frameshifts that disrupt the gene. The cell now has no working copy, so you can see what the gene was doing.
- **Knock-in**: you supply a donor template and the cell repairs by **HR** (homologous recombination), which is precise — you can insert a tag, a reporter, or a specific mutation at a chosen site.

The control that makes it an experiment. An **isogenic control** — a pair of cell lines that are genetically identical except for the single edit under study. Any difference you observe is then attributable to that edit and nothing else. This is why the technique is trustworthy: the comparison is built in.

Making the proteins separable. When you need two proteins from one construct, a **T2A** peptide (a flexible linker) lets the ribosome skip one peptide bond and produce two separate proteins from a single mRNA — for example GFP alongside your gene of interest, so you can sort or track the cells.

Moving from cell lines to patients. Cell lines are not patients. Two higher-fidelity systems close the gap:

- **Patient-derived organoids** — clinical tissue grown in 3D to mimic the architecture and physiology of the organ, used to test drug response and to carry out CRISPR/Cas9-mediated knock-in in a patient-derived system, including **colorectal cancer patient-derived organoids**.
- **Organ-on-a-chip / liver-chip** — a microfluidic device that models the liver's physiological environment for compound screening.

The comparison you must say out loud:

> *"CRISPR/Cas9 with an isogenic control is what turns a gene list into a causal claim: knock the gene out and compare against an otherwise identical cell, in a human-relevant system such as patient-derived organoids or an organ-on-a-chip."*

B2 The misconceptions to delete before you start

Delete these before reading the chains; each one will cost you a mark.

The wrong belief	Why it is wrong
"A biomarker is a test."	A biomarker is the **biological entity**; the test is the **companion diagnostic**.
"Prognostic and predictive mean the same thing."	Prognostic = outcome regardless of treatment; predictive = differential benefit from a **specific** treatment, and it needs a treatment arm to be shown.
"A good prognostic test justifies giving a drug only to some people."	It does not. That is a **predictive** claim. A prognostic test tells you who is at risk; it does not tell you who benefits.
"MammaPrint and Oncotype DX are two versions of the same test."	Different gene counts (70 vs 21), different platforms (microarray vs RT-qPCR), different risk bands (2 vs 3), and only Oncotype DX is shown to be predictive.
"Real-time PCR gives you an absolute number by itself."	Only with an **external standard curve**. ddPCR gives you an absolute number without one.
"Microarray is obsolete, so MammaPrint is invalid."	It is obsolete **for research discovery**; the 70-gene assay is standardised, validated and FDA-cleared.
"You can validate a signature in the cohort you found it in."	You cannot. You need an **independent cohort**.
"Adjacent tissue is just tumour tissue that has not yet turned cancerous."	It is histologically normal but molecularly abnormal, and its signature is independently prognostic.

The wrong belief	Why it is wrong
"A high Recurrence Score means the patient has a poor prognosis."	True — but the reason to care is that it identifies who **gains from chemotherapy**. Say both, or you have answered half the question.
"Low RS means chemotherapy is harmless."	No — the odds ratio for chemotherapy in low-RS patients **exceeded 1**, which suggests net harm.
"Sensitivity and specificity are properties of the disease."	They are properties of the **test at a chosen cut-off**; the ROC curve is how you choose it.
"Analytical validation and clinical validation are the same kind of thing."	Analytical is a **laboratory** property; clinical is a **population** property.
"Autoantibodies are a nuisance in immunoassays."	Autoantibodies **amplify** the signal and are good biomarkers. It is **anti-drug** antibodies that are the nuisance.
"CRISPR knock-out and knock-in work the same way."	Knock-out exploits error-prone **NHEJ**; knock-in uses **HR** with a donor template.
"The tumour-adjacent tissue study is a side note."	It carries his one message for the whole lecture: *"don't only look at the cancer tissue, but also the adjacent tissue."*

B3 A five-day plan from zero

Five days, roughly two hours each, in the order the lecture teaches. Each day has one output you must have at the end of it.

Day	Read	Work through	Output at the end of the day
1	§1–§2, A1, Chain 1, Chain 2	Draw the pipeline on a blank page and fill in every box. Then write one sentence on driver vs passenger that a classmate could mark as correct.	One A4 page: the pipeline, and the driver-vs-passenger sentence
2	A2 (all four traps), Chain 3, Chain 4	Write the four traps from memory on a fresh sheet, then check. Redo any you got wrong.	The four trap sentences, written out
3	A3, A4, Chain 5, Chain 6, Chain 7	Make a table with the five columns *MammaPrint / Oncotype DX / genes / platform / output*, then fill the RS bands and the trial numbers from memory.	The comparison table and the NSABP B-20 numbers
4	A5, Chain 8, Chain 9, Chain 10	Draw the validation ladder — analytic, clinical validity, clinical utility — and write the one thing that distinguishes each rung. Then list the five analytical parameters and say what each one catches.	The ladder, and the five parameters with their failure modes
5	A6, A7, A8 (self-test)	Sit the 15 questions closed book in English. Mark yourself. Re-read only the questions you got wrong, then re-read the chain they came from.	The self-test marked, with a list of what to revisit

On the last night before any exam, read Part A only. It is written to be re-readable in an hour.

Figure L7l — The five-day plan from zero



Figure L7l. The five days, the work on each, and the output you must have at the end of it, with the two drills that matter more than the reading. Figure authored at 6.5 in print width and placed 1:1 on this page.

How to read this plate: one row per day, in order - the days are ordered so that each supplies the vocabulary the next one needs. Tick the right-hand column as you go. On day 5 evening, and on the night before any exam, read Part A only.

B4 On the day

1	**Read the question twice before writing anything.** Underline the verb: *explain*, *describe*, *compare*, *calculate*, *list*. If it says **explain**, do not describe the data.
2	**Answer in English.** No marks for a Chinese answer.
3	**Lead with the discriminator.** "This is a **predictive** biomarker, because…" before anything else.
4	**One idea per sentence.** Short and clear.
5	**Show the mechanism as a chain** if the question asks how something works.
6	**Include the number** if you know it. Numbers are cheap marks.
7	**Add the limitation** if the question asks about a method or an assay.
8	**If you do not know, write what you do know** in the right frame — the frame often carries the mark even when the detail does not.

Appendix A — Where each claim in this book comes from

Everything here is traceable to the Lecture 7 recording of 7 Nov 2026 or its 42-slide deck. Claims that are **the lecturer's own words** are marked with a quotation; everything else is a faithful restatement of what he taught.

Section	Source
Front matter	the recording's own meeting toolbar clock (4:12 PM, 11/7/2026), read from the first minute of the recording; deck page 1 (title slide)
A1.1	his own words at the start of the biomarker block — *"another logical question is how we develop these biomarkers… the next few slide may be very interesting to you guys"*
A1.2	his words at the tumour-adjacent-tissue slide — *"don't only look at the cancer tissue… very clear and simple message"*
A1.3	his words at the Oncotype DX gene slide — *"don't memorize all this genes. ID is not necessary"*
A1.4, A1.5	the opening of the lecture; the standing principle is from Lectures 1 and 4
A2, Trap 1	his words at the HER2/clinical-trial slides — *"there's a two-step concept"*
A2, Trap 2	his words at the personalised-medicine slide — *"prognostic markers and also the predictive markers"*
A2, Traps 3–4	the Oncotype DX, RS and biomarker-development slides; the FDA development framework slide
A3.1	the drug-development pipeline and personalised-medicine slides; his spoken figures for trial cost and IND cost
A3.2	the EGFR T790M and droplet digital PCR slides, and the standard-curve figure
A3.3	the MammaPrint workflow, discovery and validation slides; the Oncotype DX and 21-Gene RS slides
A3.4	the Oncotype DX and chemotherapy benefit slides (NSABP B-20)
A3.5	the "RS stratifies patients by their 10-year distant recurrence" slide (Paik et al., *N Engl J Med*, 2004)
A3.6	the "Circulating microRNA for early HCC detection" slide (Liu AM et al., 2012) and the cancer-adjacent-tissue slide
A3.7	the analytical validation slides — sensitivity, specificity, dynamic range, reproducibility, matrix effect
A4	every pair is drawn from a comparison he made explicitly, on a slide or aloud
A5	method selection, built from his stated reasons for each technique
A6	the four moves and the habits follow the "explain, don't describe" correction he gave after the midterm in Lecture 6
A7	his explicit "don't memorize" and "don't be scared by all these figures" statements
B1	chain-by-chain reconstruction of the lecture in teaching order; each chain's comparison sentence is the form he uses when he compares two things aloud
Appendix C	independent check of the technical claims against published sources

Appendix B — Key figures from the lecture deck

The deck has 42 numbered pages; the recording shows several of them more than once as he returns to them. Page numbers are the deck's own.

Page	Topic
1	Title — Genomic Applications in Biomedical Research

Page	Topic
2	Overview: pathway enrichment analysis · biomarkers in clinical trials · gene signatures in breast cancer · biomarker development workflow · CRISPR/Cas9
3	Pathway enrichment analysis of omics data — the definition, and the graphical abstract
4	Pathway enrichment analysis workflow — g:Profiler, GSEA, Cytoscape, EnrichmentMap
5	Example: using PEA to identify driver genes in colorectal cancer — the network and the enrichment bars
6	Biomarkers are important to drug R&D
7	Use of biomarkers can increase the success rate of clinical trials
8	The same, with the table of trials **with** and **without** a CDX
9	Personalised medicine — population, patient sub-populations, treatment
10	EGFR T790M screen in lung cancer patients using PCR
11	Droplet digital PCR — the microfluidic partitioning and the counting
12	Gene signature tests for breast cancer — MammaPrint, Oncotype DX, ER+ / HER2−, risk of recurrence, response to chemotherapy
13	MammaPrint — the general workflow, tumour → RNA → cDNA → dye → chip → scanner → risk score
14	Discovery of the prognostic 70-gene signature (van 't Veer et al., *Nature*, 2002)
15	Validation of the 70-gene signature in an independent cohort, **n = 295** (van de Vijver et al., *N Engl J Med*, 2002)
16–17	Oncotype DX — 21 genes (16 cancer + 5 reference), RT-qPCR, target patients, sample types
18	21-Gene Recurrence Score — the five gene groups, the formula, the range and the bands
19	RS stratifies patients by their 10-year distant recurrence — the multivariate table (Paik et al., *N Engl J Med*, 2004)
20	Oncotype DX and chemotherapy benefit — NSABP B-20: purpose, 651 patients, 227 vs 424, FFPE blocks
21	Chemotherapy benefits only patients with high recurrence risk
22	Magnitude of chemotherapy benefit in patients with high RS
23	Prognostic value of cancer-adjacent tissue (2 cm from the tumour edge) in ER+ breast cancer
24	NGS studies on pharmacogenomics
25	Pharmacogenomic biomarkers in drug labelling — the FDA table, abrocitinib / CYP2C19
26	Framework for biomarker development — the FDA figure
27	Best: Biomarker, Endpoints and other Tools
28	Biomarker Development: A Multidisciplinary Team Effort
29–32	Analytical validation — sensitivity, specificity, dynamic range, reproducibility, matrix effect
33–35	Clinical validation — clinical validity, clinical utility, the ROC curve, and the microRNA HCC case
36–37	Clinical trial design — the enrichment design and the all-comer design
38–39	CRISPR/Cas9 in biomedical research; knock-out and knock-in
40–41	CRISPR/Cas9 knock-in in colorectal cancer patient-derived organoids; the liver-chip
42	Summary

Appendix C — External verification, and the corrections it produced

Everything in Parts A and B is drawn from the lecture recording. This appendix records the independent check of the technical claims against published sources. It matters here more than usual: **eight of the deck's citations are wrong or materially misleading**, one slide claim is a **reversed** finding, and one number is being read off the slide at the wrong scale. If a question ever asks you to go beyond the slide, this is the page to read.

The check was run on 8 November 2026 against primary literature and official sources. Where a source could not be machine-read (paywalls and 403s), the item is listed in C.5 as **unverified** rather than guessed.

C.1 The course

Item	Verified	Source
Institution	**The Chinese University of Hong Kong (CUHK)**, School of Biomedical Sciences, Faculty of Medicine	http://bioinfo.cuhk.edu.hk/programmes.php
Programme	MSc in Genomics and Bioinformatics; GNBF5020 *Introduction to Molecular Biology and Genetics* is a **3-unit** course in Term 1	http://bioinfo.cuhk.edu.hk/programmes.php · https://www3.sbs.cuhk.edu.hk/en/msc-in-genomics-and-bioinformatics
Class time	**Saturday 16:15–18:30**, CUHK campus	same
Medium of instruction	**English** (the course is tagged (EN) in the CUHK registry)	same
Assessment weighting	**Assignments 30 % · mid-term 30 % · final 40 %**	https://www.wfsfaa.gov.hk/tc/resources/course_search/cef_details.php?course_code=41M124753
Past paper / sample exam	**None publicly indexed.** No exam blueprint or specimen paper is published for this course.	—
The date this lecture was actually delivered	The recording's own meeting toolbar shows a live clock reading **4:12 PM, 11/7/2026**. The Lecture 6 recording's toolbar reads **10/31/2026, 4:15 PM**, advancing live to 6:28 PM.	the two recordings themselves

What this means for you. The paper structure you experience is set by the lecturer, not published — but the published **30 % mid-term** component is consistent with what he described, and the **40 % final** tells you the final paper carries the largest single share of the course grade. That is the one external fact about the exam that is actually documented.

C.2 Corrections you should know about — ordered by how likely each is to cost a mark

No.	The slide says	What the sources say	What to write in the exam
1	The "RS stratifies patients…" slide gives **HR 2.81 (1.70–4.64), P < 0.001** for the Recurrence Score, sitting in a table of per-covariate hazard ratios.	Paik et al. 2004 computed this hazard ratio "relative to an increment of 50 units" of the score, and the Ontario HTA reproduces it as "Multivariate Cox proportional HR (95 % CI) for 50-point increment in continuous RS: 2.81 (1.70–4.64), P < 0.001".	**HR 2.81 is per 50-point increment of RS, not per 1 point.** Saying "per unit" overstates the score's power by a factor of fifty. Quote it as **"HR 2.81 (1.70–4.64) per 50-point increment, P < 0.001"**.
2	The assay is "developed from **NSABP B-14** and validated with data and specimens from **NSABP B-20**".	**Inverted.** Paik 2004: the algorithm was derived from "three independent clinical studies … involving a total of 447 patients, including the tamoxifen-only group of NSABP trial B-20", and then *"the study reported here"* — the prospective validation — was run on archival **B-14** blocks: *"No samples from trial B-14 were used for prior testing or training."*	Derivation used **B-20** (plus two other cohorts, 447 patients); prospective validation was on **B-14** (668 of 675 blocks analysable).
3	The chemotherapy-benefit evidence is often cited as *"Sparano et al., NEJM, 2005"*.	**No such paper exists.** PubMed returns **0 records** for Sparano in *NEJM* 2005, and 0 for *NEJM* 2005 with "21-gene" or "recurrence score". The paper is **Paik S, Tang G, Shak S, et al. "Gene expression and benefit of chemotherapy in women with node-negative, estrogen receptor–positive breast cancer." J Clin Oncol. 2006;24(23):3726–3734. doi:10.1200/JCO.2005.04.7985**	Cite **Paik et al., J Clin Oncol 2006**. It was a **retrospective** analysis of the randomised B-20 trial: 651 patients, interaction test **P = 0.038**, large benefit in RS ≥ 31, little or none in low RS.
4	The adjacent-tissue slide is cited *"Badiat-Dekic et al., 2013"*.	There is **no author record for "Badiat-Dekic" in PubMed at all** (0 results). The paper with "Active" / "Inactive" extratumoral subtypes is **Román-Pérez E, Casbas-Hernández P, Pirone JR, et al. "Gene expression in extratumoral microenvironment predicts clinical outcome in breast cancer patients." Breast Cancer Research 2012;14:R51. doi:10.1186/bcr3152**	If a citation is asked for, the paper is **Breast Cancer Research 2012;14:R51**. The deck's citation does not exist.
5	The "inactive" adjacent signature is enriched for **anti-tumour immunity** genes, and that is why those patients do better.	**The direction is reversed in the paper.** In Román-Pérez et al., the **Active** subtype is enriched for *inflammatory response, cell movement, chemotaxis, fibrosis and macrophage recruitment*; the **Inactive** subtype is defined by *claudins, cell adhesion, cell–cell contact and epithelial differentiation*. The word "immune" appears in the paper only in the introduction.	**Exam answer: say what he taught** — the adjacent tissue carries a prognostic signature, and the favourable ("inactive") one is the one he describes as carrying anti-tumour immunity genes. **Know separately** that the paper's Active subtype is the inflammatory, macrophage-recruiting, wound-healing one. He is describing the signature's biology loosely; the paper's own axis is fibrosis-and-inflammation versus adhesion-and-differentiation.
6	The "Validation of 70-gene signature (n=295)" slide is cited *"Desmedt C et al., Int J Cancer, 2007"*.	Desmedt et al. 2007 is the **76-gene** signature, in **Clin Cancer Res** 13(11):3207–3214, n = **198** — a different assay. The **n = 295** validation of the 70-gene signature is **van de Vijver MJ et al., N Engl J Med 2002;347(25):1999–2009** (180 poor-prognosis and 115 good-prognosis patients; HR 5.1, 95 % CI 2.9–9.0 for distant metastasis).	The **295** and the **van de Vijver** citation go together. Further TRANSBIG validation: Buyse M et al., *JNCI* 2006;98:1183–1192.
7	The Oncotype DX slide lists the invasion gene as **"ST3 (MMP-11)"** and **"CATL2"**.	**ST3 is the protein alias of MMP11** (stromelysin-3), not a separate gene — writing it as a parenthetical alias merges two different genes. And the gene is **CTSL2** (cathepsin L2), not CATL2. Note also that **BCL2 sits in the oestrogen-receptor group** in the original figure although it is an apoptosis gene.	Say **"MMP11 (its protein alias is ST3) and CTSL2, both in the invasion group"**.

No.	The slide says	What the sources say	What to write in the exam
8	The FDA table shows abrocitinib as **NDA 213878**, label version **01/14/2022**.	The application is **NDA 213871** (Cibinqo, Pfizer, approved 14 Jan 2022); **213878** is a different application. The label has since been revised several times — the current DailyMed label is **version 8**. The pharmacogenomic content has also **moved**: it now sits in **2.4** (Dosage and Administration), **8.8** (Use in Specific Populations) and **12.5** (Clinical Pharmacology → Pharmacogenomics), not 2 / 8 / 12.1.	**NDA 213871**. Quote the dosing rule, which is unchanged: CYP2C19 poor metabolisers start at **50 mg once daily**, escalating to **100 mg once daily** if the response is inadequate after 12 weeks, because **AUC is increased from reduced metabolic clearance**.
9	The BEST slide is headed "Best: Biomarker, Endpoints and other Tools" and attributed to **NCI**.	**BEST** is a joint **FDA–NIH** resource — *"The U.S. Food and Drug Administration (FDA) and the National Institutes of Health (NIH) are joint sponsors"* — and its official styling is **BEST (Biomarkers, EndpointS, and other Tools)**, the capital S marking the plural. No NCI-branded "BEST" resource exists.	Attribute it to the **FDA–NIH Biomarker Working Group**, and give the NCBI Bookshelf record: **NBK326791** (https://www.ncbi.nlm.nih.gov/books/NBK326791/).
10	The lecturer's figures: **US$200,000 per patient** enrolled in a trial, and **US$1 million** to submit an IND.	Neither figure has a citable published source. The best peer-reviewed estimate of pivotal-trial cost is a **median of US$41,413 per patient** (IQR US$29,894–75,047) across the 225 pivotal trials supporting 101 FDA approvals in 2015–2017 (Moore TJ, Heyward J, Anderson G, Alexander GC, *BMJ Open* 2020;10:e038863). For the IND, the only official number nearby is **FDA's own review-cost standard for an IND application, ≈US$0.46–0.62 million** per application — a different quantity from the sponsor's cost of preparing one.	Attribute them: *"Dr. Wong's figures in the lecture are about US$200,000 per patient enrolled and about US$1 million for an IND submission."* Do not cite them as published estimates.
11	The microRNA slide is cited *"Liu AM et al., 2012"*, on a *J Mol Diagn* template.	The paper is **Liu AM, Yao T-J, Wang W, et al. "Circulating miR-15b and miR-130b in serum as potential markers for detecting hepatocellular carcinoma: a retrospective cohort study." BMJ Open 2012;2(2):e000825. doi:10.1136/bmjopen-2012-000825**. There is no *J Mol Diagn* paper of that title in PubMed.	**BMJ Open 2012;2:e000825**. Every number on the slide is correct.
12	The g:Profiler slide implies a Broad Institute tool.	**g:Profiler is a University of Tartu (BIIT) resource** and part of the **ELIXIR** infrastructure; the Broad connection is via the parallel Cytoscape/GSEA ecosystem, not ownership. Its core tool **g:GOSt** performs **over-representation analysis** against GO, KEGG, Reactome, WikiPathways, miRTarBase, TRANSFAC, CORUM and HPO.	"g:Profiler (University of Tartu / ELIXIR, https://biit.cs.ut.ee/gprofiler/)" — and describe it as **over-representation analysis (ORA)** of a gene list, not just "GSEA".
13	The ddPCR slide, with no citation.	The right primary sources for the Bio-Rad platform are **Hindson BJ et al., Anal Chem 2011;83(22):8604–8610** (high-throughput ddPCR) and **Pinheiro LB et al., Anal Chem 2012;84(2):1003–1011** (copy-number quantification, duplex assays).	Cite **Hindson 2011 / Pinheiro 2012**. The science on the slide is right: ~20,000 monodisperse 1-nL droplets per 20-µL reaction, positive/negative counting, Poisson statistics, **no standard curve**.

C.3 Claims that checked out exactly as taught

These were verified against a primary or official source and need no adjustment — state them with confidence:

- **The assay architecture.** Oncotype DX is a **quantitative TaqMan RT-PCR** assay on **fixed, paraffin-embedded tumour tissue**, measuring the mRNA of **16 cancer-related genes plus 5 reference genes**, combined by a prospectively defined algorithm into a **Recurrence Score from 0 to 100**. (Paik S, Shak S, Tang G, et al. N Engl J Med 2004;351(27):2817–2826. doi:10.1056/NEJMoa041588)
- **The five reference genes are** ACTB (β-actin), **GAPDH**, **GUS**, **RPLP0** (written *RPLPO* in the 2004 paper) and **TFRC**.

- **The risk bands, exactly as on the slide.** Paik 2004: *"low risk (recurrence score, less than 18), intermediate risk (recurrence score, 18 or higher but less than 31), and high risk (recurrence score, 31 or higher)"* — the slide's **< 18 / 18 to < 31 / 31 to 100** is the same convention.
- **The B-14 numbers.** RT-PCR succeeded in **668 of 675** available paraffin blocks; **51 %** of patients were low risk, **22 %** intermediate, **27 %** high; 10-year distant recurrence was **6.8 % (4.0–9.6)**, **14.3 % (8.3–20.3)** and **30.5 % (23.6–37.4)**, with **P < 0.001** for low against high.
- **The RS is prognostic and predictive.** The same paper established the score as an independent predictor in a multivariate Cox model, and the **B-20** analysis (Paik et al., *J Clin Oncol* 2006) showed the benefit of chemotherapy confined to the high-score group. The score was also predictive of overall survival.
- **The B-20 trial population.** *"A total of 651 patients were assessable (227 randomly assigned to tamoxifen and 424 randomly assigned to tamoxifen plus chemotherapy)"* — retrospective archival FFPE tissue.
- **The miRNA classifier — every single number.** 220 miRNAs profiled by qPCR on tumour and matched adjacent non-tumour tissue from **96** Chinese HCC patients; **six** consistently tumour-high (miR-15b, -21, -130b, -183, -224, -301); **four** carried into serum; a two-miRNA (**miR-15b + miR-130b**) classifier at a cut-off of −0.611; **15** patients sampled before and after surgery showed serum levels falling, indicating tumour origin; independent validation in **30** healthy controls, **29** chronic hepatitis B carriers and **57** HCC patients, giving **AUC 0.981**, **sensitivity 98.25 %**, **specificity 91.53 %**, **accuracy 94.83 %**.
- **MammaPrint's composition.** A **70-gene** prognostic signature; from **~25,000** genes on an inkjet-synthesised oligonucleotide array, ~5,000 were significantly regulated, **231** genes were selected as prognostic and **70** were compiled into the signature (van 't Veer LJ et al., *Nature* 2002;415:530–536). Commercialised by **Agendia**; it reports **HIGH** or **LOW** risk.
- **MammaPrint's regulatory status.** FDA-**cleared** (510(k)), not an FDA-approved drug.
- **g:Profiler, GSEA, Cytoscape and EnrichmentMap.** GSEA *"determines whether an a priori defined set of genes shows statistically significant, concordant differences between two biological states"*; cite **Subramanian A et al., PNAS 2005;102(43):15545–15550**. EnrichmentMap is a Cytoscape 3 app that *"visualizes enrichments of pathways as an enrichment map, a network representing overlaps among enriched pathways"* — nodes are gene sets, edges are mutual overlap, so redundant pathways cluster.
- **ddPCR's mechanics.** *"ddPCR measures absolute quantities by counting nucleic acid molecules encapsulated in discrete, volumetrically defined, water-in-oil droplet partitions"*; **Poisson correction without calibrators**, and **duplex assays** quantify two loci in one reaction.
- **The CYP2C19 poor-metaboliser rule.** *"the recommended dosage of CIBINQO is 50 mg once daily … If an adequate response is not achieved with CIBINQO 50 mg orally daily after 12 weeks, consider increasing dosage to 100 mg orally once daily"* — and *"in patients who are CYP2C19 poor metabolizers, the AUC of abrocitinib is increased … due to reduced metabolic clearance."*
- **The autoantibody amplification mechanism.** A tumour neoantigen is recognised as non-self, B cells clonally expand, and high-titre autoantibodies circulate — so the antibody titre is far above the neoantigen concentration itself. This is the biological reason autoantibodies make sensitive biomarkers, and it is the opposite of what an ADA does.
- **The ADA interference mechanism.** A therapeutic antibody is foreign protein, so the host raises anti-drug antibodies, which bind the drug in circulation — neutralising efficacy and competitively blocking the assay's capture/detection antibodies in a pharmacokinetic sandwich ELISA.
- **The clinical-trial phase definitions and participant numbers.** Preclinical → IND → **Phase 1 (20–80)**: safety and dose → **Phase 2 (100–300)**: efficacy and side effects → **Phase 3 (1,000–3,000)**: confirm efficacy against standard care → FDA review and approval → **Phase 4 (1,000+)**: post-marketing surveillance.

C.4 Where the literature and the lecture use different words for the same thing

Three places where the vocabulary can cost you a mark if you use the wrong synonym.

Concept	Standard names	The lecturer's words	Same thing?
Screening a trial population by a biomarker	**enrichment design** · enrichment strategy · biomarker-driven selection	**enrichment design**	[yes] yes
Enrolling everyone and correlating the biomarker afterwards	**all-comer design** · all-comers trial · unselected design	**all-comer design**	[yes] yes
The assay that measures the biomarker	**companion diagnostic (CDx)**	**CDX**	[yes] yes

His convention matches the standard one, so you may use either vocabulary. The mark is lost by *mis-describing the design*, not by choosing a different name. Always lead with the design — "an **enrichment** design, where the biomarker is used as an initial test to screen the patients" — and then give the reason.

C.5 What could not be verified, and is therefore not claimed in this book

- **The individual covariate hazard ratios on the "RS stratifies patients…" slide** — age at surgery 0.76 (0.50–1.18), clinical tumour size 1.19 (0.81–1.76), moderately differentiated grade 1.55 (0.85–2.81), poorly differentiated grade 3.34 (1.79–6.26), HER2 amplification 0.51 (0.26–1.02), ER protein 50–99 fmol/mg 0.75 (0.43–1.31), 100–199 fmol/mg 0.90 (0.52–1.58), ≥ 200 fmol/mg 1.02 (0.58–1.70). Every accessible copy of Paik 2004 renders that table without its numeric cells, and the paper's own footnote does confirm **how** the covariates were coded — age as a binary variable at 50 years, tumour size as a binary variable at 2 cm, grade as poorly versus well/moderately differentiated, HER2 amplification as a binary variable, and ER protein as an ordinal variable with a 10–49 fmol/mg baseline — and that **only the Recurrence Score and poor tumour grade were independent predictors**. **The numbers on the slide are reproduced in A3.5 because that is what he taught, but check them by hand against the paper before quoting them as sourced.**
- The exact array designation **"Hu25K"** for the MammaPrint discovery array: the ~25,000-gene inkjet-oligonucleotide array is confirmed; that particular name was not.
- The **"1,000 or more"** figure for Phase 4 and the **FDA-hosted** version of the phase-participant table: the 20–80 / 100–300 / 1,000–3,000 figures were confirmed from the NIH page and from FDA-republished patient materials, but FDA's own phases page could not be opened from this environment.
- No published **past paper, sample exam or exam blueprint** for GNBF5020 is publicly available. Every prediction in Part A is therefore drawn from the lecturer's own statements in this recording, not from a published paper.
- The **current guideline status** of MammaPrint and the present-day commercial details of both assays were not confirmed beyond the sources cited above; only their composition, platform, indication and clearance are stated here.

Verified against the sources listed above on 8 November 2026. Where the lecture and the literature differ, this book says so explicitly — because "seeing is believing" is his instruction to you, and the best way to honour it is to know which number he will want on the page and which one is actually right.